# 13.11 What Values Make an Equation or Inequality True?

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmark | MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false. | Learning Target | • I can determine if a given integer value makes an equation or inequality true. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.5.AR.2.3 Determine and explain whether an equation involving any of the four operations is true or false. | MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically. |

| | |
|---|---|
| Essential Questions | • How are solutions to equations and inequalities the same? How are they different?<br>• What is the difference between a strict inequality and an inequality?<br>• How can you use mental math to determine whether inequalities are true or false? |
| Lesson Narrative | In this lesson, students determine if equations and inequalities are true by substituting values into the equation or inequality. Students learn the meaning of the equal sign and the inequality symbols. Students use mental math strategies to determine whether inequalities are true or false. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 13.11.1 Warm-Up (~5 min.) | Day at Work – Film Composer *(SE p. 321)* | 13.11.4 Guided Instruction (15 min.) | Determine if inequalities are true *(SE p. 323)* |
| | 13.11.2 Guided Instruction (10 min.) | Determine if equations are true for a given value *(SE p. 321)* | 13.11.5 Your Turn! (5 min.) | Practice using substitution to determine if inequalities are true for given values *(SE p. 326)* |
| | 13.11.3 Guided Practice (5 min.) | Practice determining if equations are true *(SE p. 323)* | 13.11.6 Wrap-Up (~5 min.) | Self-assess understanding of the learning target *(SE p. 326)* |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | Key Terms:<br>• Equation<br>• Solution<br>• Inequality<br>• Strict inequality<br><br>Additional Terms: N/A | • Career Profile video | • The 13.11.1 Warm-Up video can be found in the online Teacher Area to present for students before they answer questions in the Warm-Up. |

| | Common Misconceptions | Things to Consider |
|---|---|---|
| Teacher Tips | • Students may believe the symbol < can only mean or represent "less than," even though an expression like $x < 3$ can be read as "3 is greater than $x$."<br>• Students may struggle with operations with integers. | • Avoid using common strategies like alligator inequalities that perpetuate misconceptions. Such shortcut methods that are not mathematically based, often confuse students. Instead, provide space and opportunity for students to articulate the meanings of symbols in their own words and give examples.<br>• Consider giving students number lines to help them evaluate if the inequality is true or false.<br>• If time is an issue, then consider assigning 13.11.5 Your Turn! for homework. |
| | Literacy and Language Routines | Mathematical Thinking and Reasoning (MTR) Standard |
| | **True or False**<br>13.11.4 Guided Instruction provides an opportunity for students to consider mental math strategies for determining whether inequalities are true. | **MA.K12.MTR 6.1: Reasonableness**<br>Throughout the 13.11.2 Guided Instruction, 13.11.3 Guided Practice, 13.11.4 Guided Instruction, and 13.11.5 Your Turn!, students assess the reasonableness of their responses. Students are encouraged to use mental math to determine whether inequalities are true, encouraging mathematical reasoning. |
| Differentiate Instruction | Consider allowing students who need additional scaffolding access to the $6^{th}$ Grade On-Ramp video titled True or False Equations. | |
| Lesson Notes: | | |

# 13.11.1 Warm-Up: Day at Work – Film Composer

***Component Overview:*** *Play the "Day at Work: Film Composer" video for students. Students then reflect on the video and how the composer's skills apply to mathematics.*

## 13.11.1 Warm-Up: Day at Work – Film Composer

Shohan Cagle is a film composer. He writes music for commercials and soundtracks for TV shows and movies.

1. What did you notice about what he does to write a soundtrack?
   *Answers will vary. May include watching the entire movie/show. May include talking to the producer of the movie/show.*

2. Shohan said that most people say they don't normally hear the soundtrack. Name a TV show or a movie whose soundtrack you remember.
   *Answers will vary. The Wizard of Oz. Elf.*

3. Shohan mentioned that one of his challenges is a short deadline. He said that his strategy to get his work done is "to sit and focus and get in the right mindset". How can you use Shohan's strategy?
   *Answers will vary. Before I start my assignment, I look over my notes. Before I start my assignment, I turn off all distractions around me.*

*Students may not be familiar with what a soundtrack is. Provide students with examples of soundtracks to familiarize them with the word. Challenge students to research how the careers they are interested in pursuing use math knowledge and skills.*

# 13.11.2 Guided Instruction: Determining Values That Make an Equation True or False

***Component Overview:*** *Students begin by reviewing the definition of equation. Students use Guess, Check, and Revise to determine the number that produces the same value for the expression on the left and right side of the equal sign. This leads students to review the formal definition of a solution. Finally, students substitute values into equations to determine if the values are solutions to the equations.*

## 13.11.2 Guided Instruction: Determining Values That Make an Equation True or False

An ***equation*** is a mathematical relation statement where two equivalent expressions and values are separated by an equal sign.

1. In the equation $4x+5=8x-23$, what are the two expressions?
   $4x+5$
   $8x-23$

2. Work with your partner to guess a value for $x$ and then substitute it into both expressions. A high and a low guess have been provided for you in the table.

| $x$ | $4x+5$ | $8x-23$ | Are the expressions equal? |
|---|---|---|---|
| $-3$ | $4(-3)+5=-7$ | $8(-3)-23=-47$ | No |
| $14$ | $4(14)+5=61$ | $8(14)-23=89$ | No |
| | *Answers* | *will* | *vary* |
| $10$ | $4(10)+5=45$ | $8(10)-23=57$ | *No* |
| $7$ | $4(7)+5=33$ | $8(7)-23=33$ | *Yes* |
| | | | |
| | | | |

For an equation to be **true**, both sides of the equation must have the same value. For example, the equation $78-56=2\cdot 11$ is true because both expressions, $78-56$ and $2\cdot 11$, are equal to $22$.

The ***solution*** of an equation is the value(s) that, when substituted in for the variable(s), makes the equation true.

*This component is intentionally scaffolded to build conceptual understanding of equations and solutions. After reviewing the definitions, students apply the definitions in the subsequent exercises. Check for student understanding as you circulate while students work. Debrief the activity, asking students to share their work.*

*Consider an inquiry-based approach to verifying solutions algebraically using substitution.*

- *Is $-3$ a solution to the equation? How do you know?*
- *How many solutions did the equation have?*

# 13.11.2 Guided Instruction: Determining Values That Make an Equation True or False (continued)

3. Based on your guesses, what is the solution to the equation $4x + 5 = 8x - 23$?
   $x = 7$

4. Use substitution to determine if the equation $2x + 4x + 7 = 19$ is true for each value.

| $x$ | $2x + 4x + 7 = 19$ | Is the equation true? |
|---|---|---|
| $-1$ | $2(-1) + 4(-1) + 7 = 19$<br>$-2 - 4 + 7 = 19$<br>$1 \neq 19$ | *No* |
| $0$ | $2(0) + 4(0) + 7 = 19$<br>$0 + 0 + 7 = 19$<br>$7 \neq 19$ | *No* |
| $2$ | $2(2) + 4(2) + 7 = 19$<br>$4 + 8 + 7 = 19$<br>$19 = 19$ | *Yes* |

5. The solution to the equation $2x + 4x + 7 = 19$ is $x = 2$ because when this value is substituted for $x$, the equation is *true*.

*An auditory entry point is provided through class and partner discussion. Consider providing a visual entry point by having students use colored pencils to substitute values so they can keep track of substitutions.*

*Students may think that all equations will only have one solution. In Grade 7 Accelerated, students find solutions to $x^2 = p$, which as two solutions. If students think that equations will always only have one solution point ahead to what students will learn in Grade 7 Accelerated and beyond.*

# 13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

***Component Overview:*** *Students substitute integers into equations to determine which values make the equations true. Reinforce that students should show their work. If students have an incorrect answer, they can retrace the steps in their work to identify their mistakes.*

## 13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

From the specified set of integers, determine which, if any, make the equation true. Show your work.

1. $18 + 7w = 10w - 9 \quad \{-9, -2, 4, 9, 11\}$

$18 + 7(-9) = 10(-9) - 9$
$18 - 63 = -90 - 9$
$-45 \neq -99$
*Not True*

$18 + 7(-2) = 10(-2) - 9$
$18 - 14 = -20 - 9$
$4 \neq -29$
*Not True*

$18 + 7(4) = 10(4) - 9$
$18 + 28 = 40 - 9$
$46 \neq 31$
*Not True*

$18 + 7(9) = 10(9) - 9$
$18 + 63 = 90 - 9$
$81 = 81$
*True*

$18 + 7(11) = 10(11) - 9$
$18 + 77 = 110 - 9$
$95 \neq 101$
*Not True*

2. $-2(3 - p) = 4p + 10 \quad \{-8, -3, 0, 10\}$

$-2(3 - (-8)) = 4(-8) + 10$
$-2(11) = -32 + 10$
$-22 = -22$
*True*

$-2(3 - 0) = 4(0) + 10$
$-2(3) = 0 + 10$
$-6 \neq 10$
*False*

$-2(3 - (-3)) = 4(-3) + 10$
$-2(6) = -12 + 10$
$-12 \neq -2$
*False*

$-2(3 - 10) = 4(10) + 10$
$-2(-7) = 50$
$14 \neq 50$
*False*

*Consider selecting and assigning some (not all) of the given values to students. This will give students more time to work on each equation.*

*Use probing questions to encourage students to use mental math strategies and improve their fluency:*
*Do you always have to determine the value for each side of the equation to determine if the equation is true? Why or why not?*

*What strategies do you use to determining the value that make the equation true?*

# 13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

***Component Overview:*** *Students review the definitions of inequality and strict inequality. Students use substitution to evaluate statements and determine if each inequality is true for the value of $x$. They also analyze a student's work to see how the student used mental math to determine if an inequality was true.*

## 13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

A ***strict inequality*** is a mathematical relationship between two expressions that are not equal. A strict inequality uses the symbols, $<$, $>$, or $\neq$.

An ***inequality*** is a mathematical relationship between two expressions that are not equal ($<$, $>$, or $\neq$) or are not always equal ($\leq$ or $\geq$).

1. Use substitution to evaluate the mathematical statement. Then determine if each inequality is true for the value of $x$.

| $x$ | **Strict Inequality** $9x-5<13$ | **Inequality** $9x-5\leq 13$ | **Which inequality is true?** |
|---|---|---|---|
| 5 | $9(5)-5<13$ <br> $40 \nless 13$ | $9(5)-5\leq 13$ <br> $40 \nleq 13$ | *Neither* |
| 2 | $9(2)-5<13$ <br> $13 \nless 13$ | $9(2)-5\leq$ <br> $13\leq 13$ | $\leq$ *Inequality* |
| $-3$ | $9(-3)-5<13$ <br> $-32<13$ | $9(-3)-5\leq 13$ <br> $-32\leq 13$ | *Both* |

2. Answer the following.

   a. The inequality $x<5$, means any number that is
   - smaller
   - larger

   than 5 will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.
   *Answers will vary.* $-\frac{3}{4}, -0.5, \frac{3}{2}, 3.25$

*Initially, students may struggle to understand the difference between inequalities and equations or may not understand how to read the symbols correctly. Students may also struggle to conceptually understand the difference between an inequality and a strict inequality. Ask guiding questions to help students make the conceptual connections:*

- *Do inequalities always express relationships between expressions that are not equal? Why or why not?*
- *What is the difference between a strict inequality and an inequality?*

*Utilize number lines as an instructional strategy for comparing values.*

# 13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

3. Answer the following.

   a. The inequality, $-11 \geq x$, means any number that is
      - smaller than or equal to
      - larger than or equal to

      $-11$ will make the inequality true.

   b. List four numbers that are not integers that make the inequality true.
      *Answers will vary.* $-\frac{83}{5}, -15.5, -\frac{64}{5}, -12.75$

4. Use substitution to determine if the inequality $16 \geq 7m - 5$ is true for each value specified for $m$.

| $m$ | $16 \geq 7m - 5$ | Is the inequality true? |
|---|---|---|
| $-2$ | $16 \geq 7(-2) - 5$<br>$16 \geq -14 - 5$<br>$16 \geq -19$ | Yes |
| $0$ | $16 \geq 7(0) - 5$<br>$16 \geq -5$ | *Yes* |
| $3$ | $16 \geq 7(3) - 5$<br>$16 \geq 21 - 5$<br>$16 \geq 16$ | *Yes* |
| $6$ | $16 \geq 7(6) - 5$<br>$16 \geq 42 - 5$<br>$16 \not\geq 37$ | *No* |

*True or False*
*Provide an opportunity for students to consider mental math strategies for determining whether inequalities are true.*

# 13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

5. Answer the following.

   a. Determine the integer(s) from the specified set that make the inequality **true**. Show your work.

   $-x+3+5x \geq -8x-7 \quad \{-12,-5,5,16\}$

   $-(-12)+3+5(-12) \geq -8(-12)-7$
   $12+3-60 \geq 96-7$
   $15-60 \geq 89$
   $-45 \geq 89$
   *False*

   $-(-5)+3+5(-5) \geq -8(-5)-7$
   $5+3-25 \geq 40-7$
   $-17 \geq 33$
   *False*

   $-(5)+3+5(5) \geq -8(5)-7$
   $-5+3+25 \geq -47$
   $23 \geq -47$
   *True*

   $-16+3+5(16) \geq -8(16)-7$
   $-16+3+80 \geq -128-7$
   $67 \geq -135$
   *True*

   b. When Diego substituted $16$ in $-x+3+5x \geq -8x-7$, he stopped after writing $-16+3+5(16) \geq -8(16)-7$ because he said he knew it was true. When his teacher asked him how he knew, he said it was because the left side would be a positive number and the right side would be a negative number.

   With your partner, discuss how Diego knew the left side of the inequality would be a positive number and the right side of the inequality would be a negative number.
   $-16+3+5(16)$
   *Since* $5(16)+3 > -16$ *the left side is positive.*

   $-8(16)-7$
   *Both values in this expression are negative, therefore their sum is a negative number.*

*As students discuss Diego's response, use guiding questions to connect their work in the previous question to Diego's work.*

- *How did Diego determine if the inequality was true? Do you agree with this method?*
- *How did you use mental math to evaluate whether inequalities were true? How is this similar to Diego's strategy?*

# 13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

6. Determine which values from the specified set make the inequality **false**. Show your work.

   $12 < 2(13y - 1) \quad \{-14, 0, 1, 9\}$

   $$12 < 2(13(-14) - 1)$$
   $$12 < 2(-182 - 1)$$
   $$12 < 2(-183)$$
   $$12 < -366$$
   *False*

   $$12 < 2(13(0) - 1)$$
   $$12 < 2(0 - 1)$$
   $$12 < 2(-1)$$
   $$12 < -2$$
   *False*

   $$12 < 2(13(1) - 1)$$
   $$12 < 2(13 - 1)$$
   $$12 < 2(12)$$
   $$12 < 24$$
   *True*

   $$12 < 2(13(9) - 1)$$
   $$12 < 2(117 - 1)$$
   $$12 < 2(116)$$
   $$12 < 232$$
   *True*

   The values from the specified set that make the inequality $12 < 2(13y - 1)$, **false** are $-14$ *and* $0$.

*Ensure students read the directions for question 6 carefully, as they are finding values that make the inequality false. This is not a misconception so much as a potential error that students may make if they do not read carefully.*

# 13.11.5 Your Turn!

***Component Overview:*** *Students independently use substitution to determine if inequalities are true or false for given integer values. Circulate while students work to monitor students' progress, support struggling students, and determine which students you will ask to share during the debrief.*

## 13.11.5 Your Turn!

1. Use substitution to determine if the inequality $86 \geq 17a - 8a - 23$ is true or false for each integer value specified for $a$.

| $a$ | $86 \geq 17a - 8a - 23$ | True or False |
|---|---|---|
| $^{-}8$ | $86 \geq 17(^{-}8) - 8(^{-}8) - 23$<br>$86 \geq -136 + 64 - 23$<br>$86 \geq {}^{-}95$ | *True* |
| $0$ | $86 \geq 17(0) - 8(0) - 23$<br>$86 \geq {}^{-}23$ | *True* |
| $7$ | $86 \geq 17(7) - 8(7) - 23$<br>$86 \geq 119 - 56 - 23$<br>$86 \geq 40$ | *True* |
| $12$ | $86 \geq 17(12) - 8(12) - 23$<br>$86 \geq 204 - 96 - 23$<br>$86 \geq 85$ | *True* |
| $21$ | $86 \geq 17(21) - 8(21) - 23$<br>$86 \geq 357 - 168 - 23$<br>$86 \geq 166$ | *False* |

*Consider assigning students a smaller subset of values to substitute into the inequalities to give them more time to evaluate each inequality and check their work.*

*For an extension, challenge students with inequalities or equations that have no solution or infinite solutions. For example:* $4(7-3x) = 5x - 17x + 28$, $2(5x-7) = 5(2x-1)$, $8(3x+7) > 6(4x-3)$, $-2(5-6x) \leq 4(3x-11)$.

# 13.11.6 Wrap-Up: Reflection

***Component Overview:*** *Students self-assess, identifying one question they have and any topics they are unsure about after the lesson. Students also brainstorm a plan for filling the gap in their understanding of the lesson's learning targets.*

### 13.11.6 Wrap-Up: Reflection

1. One question I still have after today's lesson is …
   *Answers will vary.*
   ______________________________________________
   ______________________________________________
   ______________________________________________?

2. Today I am still unsure about …
   *Answers will vary.*
   ______________________________________________
   ______________________________________________
   ______________________________________________.

3. To fill in this gap I intend to …
   *Answers will vary.*
   ______________________________________________
   ______________________________________________
   ______________________________________________.

*This 13.11.6 Wrap-Up requires a written explanation. Consider allowing students to explain their answers orally or record their responses using an app like FlipGrid.*